

The Base-Rate Trap — Facilitator Guide

Day 2 · The Eval Gap · 50 min · this studio replaces the lecture

How to run it (the math is the lecture)

- **0–5 min:** set the scenario. Ask for a show of hands: “TPR is 90% — good judge or bad judge?” Most say good. Note it.
- **5–15 min:** pods build the matrix. Don’t give the column split — make them derive 1,000 / 49,000 from the base rate themselves; that’s the whole lesson.
- **15–21 min:** compute PPV and the miss count. Put one pod’s matrix on the whiteboard and reconcile.
- **21–30 min:** effective sensitivity + the flooding twist. Re-poll the “good judge?” question.
- **30–50 min:** debrief and the “zero flags” sentence; flows straight into the 10:10 steelman-swap. Hand out gold last.

Questions to ask while circulating

- “You have 50,000 alerts. How many are truly suspicious? Write that number first.”
- “Your judge is 90% accurate. Of everything it escalates, what fraction is real? Guess, then compute.”
- “Where do the 100 missed cases go? Who looks at that pile?”
- “Does flooding the queue make the human reviewer better or worse?”
- “If the agent flagged nothing all month, would you celebrate?”

Misconceptions to watch for

- **“High TPR / accuracy = good system.”** The entire studio. The PPV (27%) lands the gap viscerally.
- **Confusing PPV with TPR.** They are different conditionals; the base rate is what separates them.
- **“99.8% NPV is great.”** True and useless — it hides 100 absolute misses a month.
- **“A better judge fixes it.”** A better judge helps, but at a 2% base rate PPV stays modest; the fix is also the human layer and an audit stream.
- **Treating vendor TPR/FPR as fact.** Flag that these must be measured on a labeled gold set.

Debrief — land this

Re-show the opening hand-count against the PPV. **Punchline:** the base rate, not the judge, sets the ceiling — a strong judge over rare events both *floods* the humans and *silently misses*, and the two failures compound. “Zero flags” at a low base rate is the absence of measurement, not the absence of failure. That is why an LLM-as-judge needs a base rate, a PPV, an effective sensitivity, and a random audit stream — not an accuracy number.

How it connects

This is ClearList’s home module and the seed of its curveball (full auto-disposition with no sampling). The effective-sensitivity idea (judge × reviewer) returns in the Day-2 HITL module as the review-queue problem, and the “random audit stream” is one of the calibration substrates from the deployment-gates material.

Illustrative for the exercise only. The numbers are ClearList’s stated figures; real judge performance must be measured on a labeled gold set, not assumed. Nothing here is legal or regulatory advice.